

“Middle Out” Design Methodology

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This Seminar Will...

- Describe two design methodologies:
 - Pure model first flow
 - Applying segments of model first within a traditional design flow
- Look at the pros and cons with the latter
- Apply ARM technology to make this “middle out” flow possible

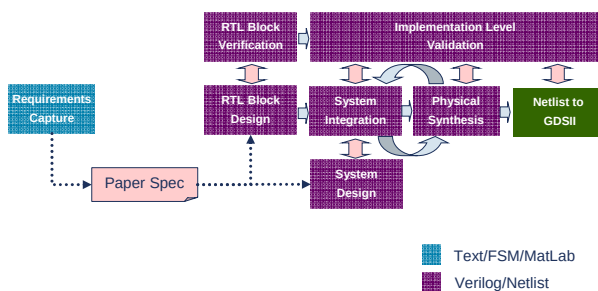


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What Do You Do Today?

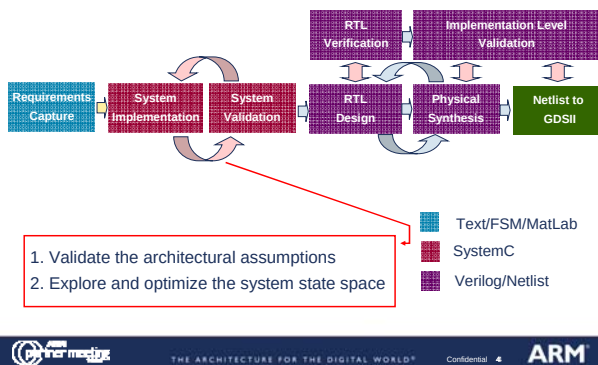


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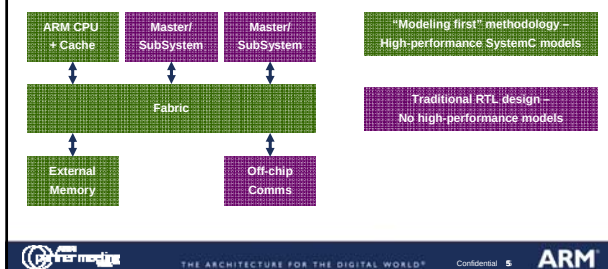


...What Would You Like to be Doing?

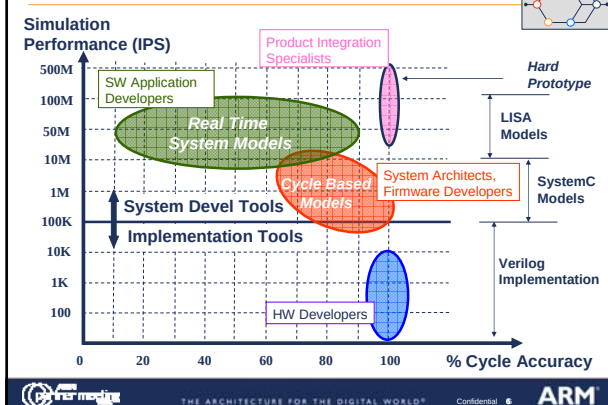


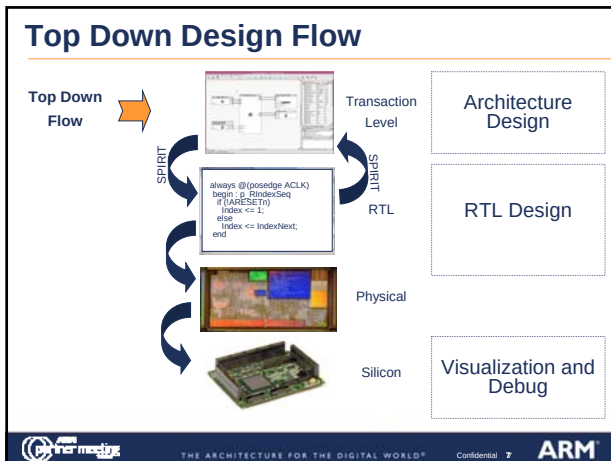
....What are the Limitations?

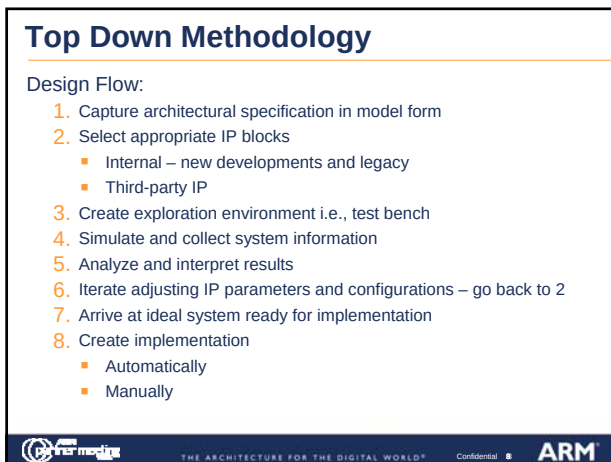
- Lack of models
- Lack of project slack
- Lack of experience in ESL

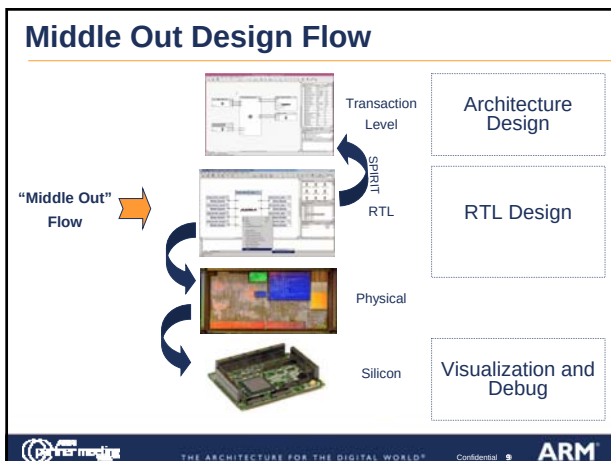


The Target Platform Spectrum









Middle Out Methodology

Design Flow:

1. Enter from predefined specification
2. Move directly to RTL generation
3. Continue with hardware implementation
4. In parallel move segments of system into modeling world
5. Perform focused modeling and analysis
6. Ripple findings through to implementations



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ARM

Enter the RealView® SoC Designer and AMBA® Designer Tools



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ARM

The RealView SoC Designer Tool

- Electronic System Level (ESL) design tool
- Rapid SoC prototyping for:
 - Architectural analysis
 - Pre-silicon software development
- Highlights:
 - Tools and utilities for component and system model creation
 - Simulator with extensive debug and profiling visibility
 - Accompanying model libraries
 - Part of a comprehensive concept to integration RealView solution



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ARM

What is the AMBA Designer Tool?

- The means to move from a specification to implementation
- Supports top down (ESL) and middle out design
- Contains ARM ESL technology
- The Fabric IP **Ecosystem**
 - Common entry point for all PrimeCell® IP
 - Explore IP configurations and fabric using ESL
 - Generates optimized RTL
 - Delivers IP in SPIRIT packaged form
- End-to-end IP optimization

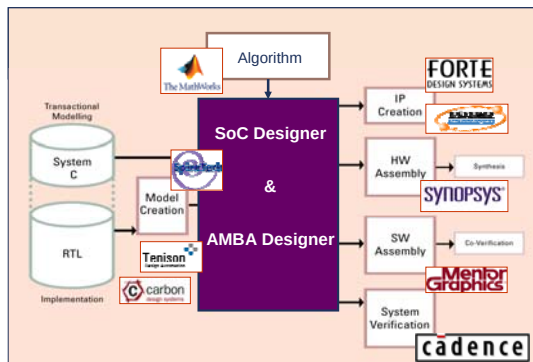


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Design-Flow Partnerships



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Optimizing The Theory

Providing a full spectrum solution to bridge between a traditional RTL design entry point to full model first methodology

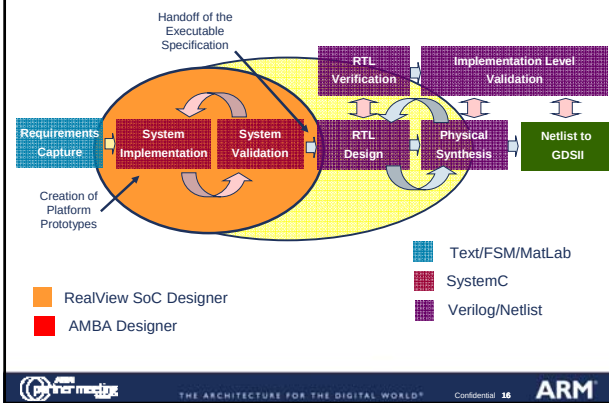


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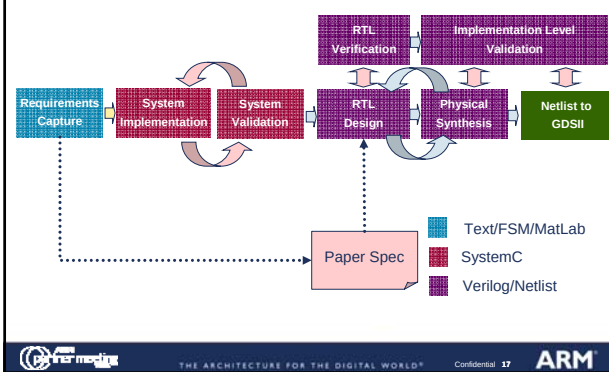
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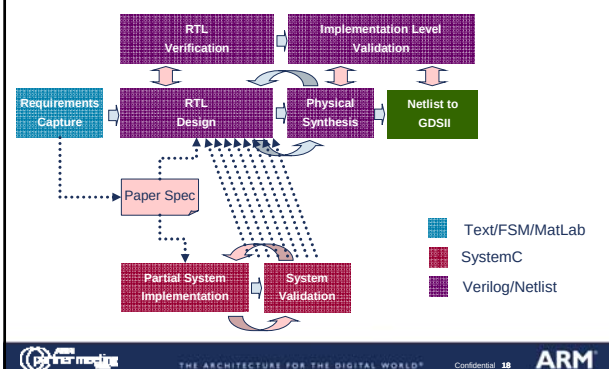
In a True Top Down Flow...

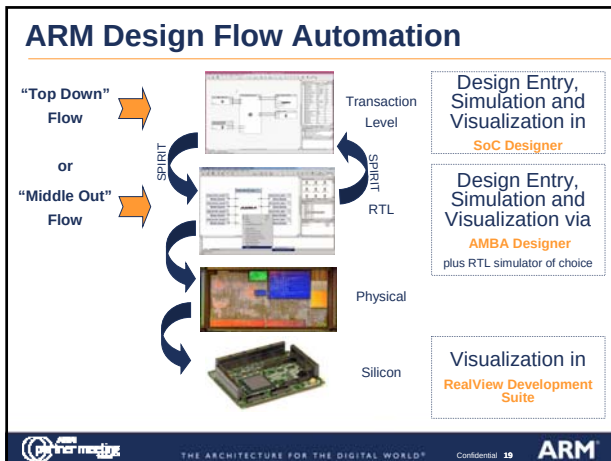


Looking at the Steps



Looking at the Steps





Putting Theory Into Practice

Applying the ARM ESL and IP Exploration and Delivery Technology

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Let's Pick an Application

- Application
 - Modern 3GSM smart phone
 - Motion video capture and playback
 - Voice and high-speed data communication
 - Enhanced audio support (MP3)
- Project Requirements
 - Flexible high-performance processor platform
 - Proactive power management
 - Minimal development time
 - Differentiating application software

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8 Steps to IP Delivery

- Step 1: Identify critical system segments
- Step 2: Pull system together
- Step 3: Best estimate system configuration
- Step 4a: IP delivery for hardware design team
- Step 4b: Data flow analysis
- Step 5: Detailed system simulation and exploration
- Step 6: Optimization
- Step 7: Final IP delivery
 - RTL + SPIRIT generation



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Rough Platform Requirements

- High-performance processing
 - Motion video and audio encoding and decoding
 - 3G base band modem
- Built-in security
 - Secure processing for e-transactions
- Intelligent power management
 - Minimal power consumption in active and standby modes
- Application software acceleration
 - Java-based applications

**ARM11™
Processor +
AMBA 3 AXI™
Interconnect +
Cache**

**TrustZone®
security
technology**

**Intelligent
Energy
Manager
technology**

**Jazelle® Java
acceleration**



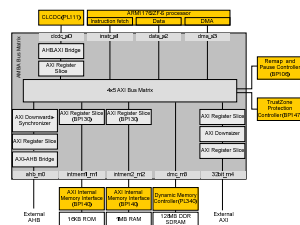
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Pulling the System Together, Quickly

- System comprised of:
 - ARM11 processor
 - AMBA 3 AXI interconnect
 - Level 2 cache
 - Custom hardware M/S
 - Memory controllers
 - Off-chip memory



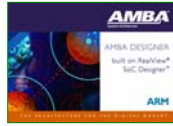
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IP Selection and Configuration

- Select IP – “Plug-and-Play”
 - Interconnect, DMA, static memory controller etc.
- Set parameters
 - Data width, protocol etc.
 - Timing, security, area etc.
- Configure requirements
 - Area, performance etc.
 - QoS, arbitration etc.



Delivers a fast cycle accurate model of the IP!



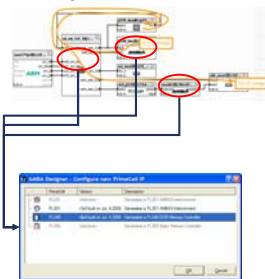
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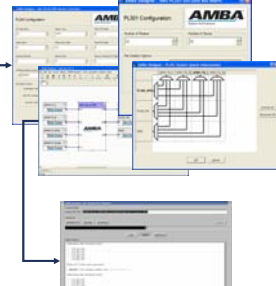


Enabling Parallel RTL Development

Identify PrimeCell IP



Configure PrimeCell IP



Generate PrimeCell™ RTL



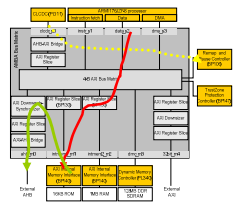
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System Data Flow

- System interconnect is
 - Central to on-chip communication
 - Critical to system performance
- Memory controllers
 - Manage the bridge between the interconnect and memory devices
 - Are often the bottleneck in system performance
- Level 2 cache
 - Acts as a bridge between the processor and the memory subsystem
 - Can have dramatic effects on system performance and power
- So, three areas to focus on:
 - Cache decision
 - Interconnect selection
 - Memory controller selection



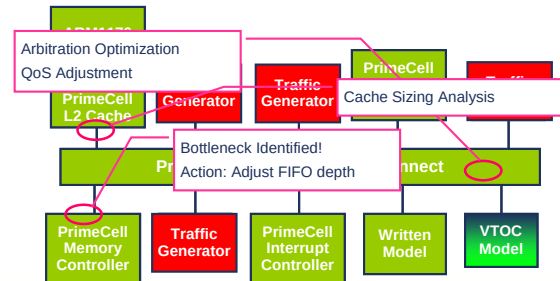
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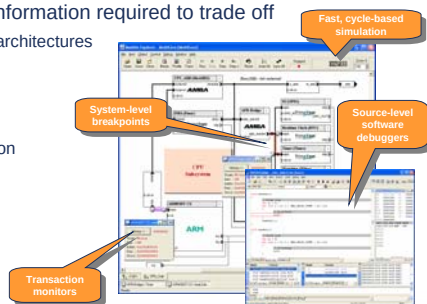
Understanding System Data Flow

- How do we understand the data flow characteristics?
- ...and make informed configuration choices?



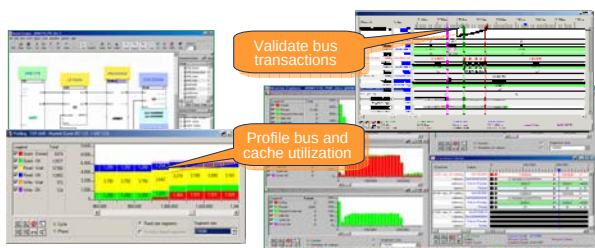
System Exploration

- Validate Fabric performance within target system
- Explore traffic scenarios
- Provides the information required to trade off
 - Interconnect architectures
 - Caches
 - Memories
 - Arbitration
 - Timing isolation
 - Etc. etc.



Data Analysis and IP Optimization

- System profiling and waveform viewing assist decision making
- Running "what if" analysis yields best overall subsystem implementation
- Alter IP configuration to provide best fit for target system



Implementation

- Custom RTL creation
- Fixed IP integration
 - RTL level IP
 - Hard IP macro with RTL simulation model
- Configurable IP integration
 - RTL delivered from golden model



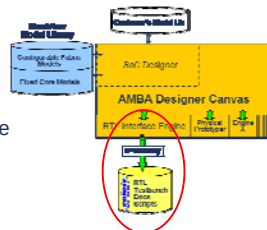
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PrimeCell RTL IP Generation

- IP Exported as SPIRIT 1.1 bundle
 - Highly optimised
 - Cleanly packaged
- PL301 is the IP delivery engine
 - Synthesizable RTL IP Package
 - Verification Suite
 - EDA Tool Scripts
- Scope for SPIRIT 2.0 Extension
 - For example – extended verification
 - Transactor support



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From ESL to Silicon Ready RTL



Ecosystem – AMBA Designer

- Next-generation methodology
- Idea capture, simulation, analysis

IP – PrimeCell Peripherals

- Powerful configurable portfolio
- Exactly meets your needs

Technology – AMBA 3 AXI

- Powerful AMBA AXI protocol
- Highest performance solution



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Summary

1. Embracing a partial modeling methodology enables
 - Rapid architectural modeling of segments
 - Fast convergence on the right IP
 - System optimization
2. Configurability in the IP provides enormous flexibility
3. Intuitive flow for system architect
4. Single ESL to RTL environment



Designer chooses performance required for the application



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